Development of a Hybrid Algorithm for Temporal Normalization of Polar-Orbiting Satellite-Derived Land Surface Temperature

Wenhui Du, Zhihao Qin, Jinlong Fan, Pei Leng, *Senior Member, IEEE*, and Zhao-Liang Li, *Fellow, IEEE*

***Abstract*— Land surface temperature (LST) is crucial in many global and regional scientific studies and applications. The observation time differs along the scan line due to the intrinsic scanning characteristics of instruments, making satellite-derived LSTs incomparable. Although many algorithms have been developed to address this issue, they have many limitations and uncertainties in application. On the basis of the temporal evolution of clear-sky LST, this study proposed a hybrid and practical algorithm with good applicability for the temporal normalization of satellite-derived LSTs. The proposed algorithm was applied based on Aqua Moderate-Resolution Imaging Spectroradiometer (MODIS) data across the contiguous United States (CONUS) in 2020 and mainly validated by cross-comparisons with the Geostationary Operational Environmental Satellite R-Series 16 (GOES-R16) Advanced Baseline Imager (ABI) LST product over each season and various land cover (LC) types. The normalized MODIS LSTs had a superior agreement with the GOES-R16 LSTs. Especially for the temporal differences (original observation time minus the reference time) between 0.5 and 1.0 h, the root-mean-square error (RMSE) and bias of the normalized LSTs were improved by 0.32–1.03 K and 0.30–1.27 K, respectively. These results demonstrate that the proposed hybrid method has advanced potential for the temporal normalization of polar-orbiting satellite LST.**

***Index Terms*— Land surface temperature (LST), machine learning, polar-orbiting data, temporal normalization.**

I. Introduction

LAND surface temperature (LST) is a comprehensive representation of water and heat balance at the interface between the atmosphere and the surface [1], [2]. In recent decades, remote sensing has become a very efficient approach to obtaining LST from local to global scales [3], [4], [5], [6], [7], [8]. Many satellite-derived LST products have been released. Specifically, the LST products generated from the Moderate-Resolution Imaging Spectroradiometer (MODIS) have been extensively applied in many scientific studies [9], [10], [11], [12], [13], such as in research on evapotranspiration [14], the urban thermal environment [15], agricultural drought [16], [17], climate change [18], soil moisture [19], hydrology [20], and agricultural productivity [21], [22].

Due to the intrinsic designed observation characteristics and large field of view of sensors onboard polar-orbiting satellites, the observation time of different pixels across the scan lines on the same image or of the same pixel on different date-observed images in one revisit period varies greatly. Taking MODIS Terra/Aqua and Fengyun-3-D Medium Resolution Spectral Imager II (MERSI-II) sensors as examples, their overall design scan widths are about 2330 and 3000 km, respectively. Also, these observation temporal differences in local solar time of east–west pixels along their scan lines can be up to approximately 2 h [4], [23], [24]. Consequently, these differences could lead to temporal in-comparability of the corresponding LSTs among the pixels within the same image. The application limitations of the time effect of LST are not only reflected in the same scene image but also impact the application of different scene images captured by the same sensor and even those taken by different sensors. For example, in agricultural drought inversion research, the stitching of multiscene data within a region (such as at the country-wide or continental scales) and the multiple-day synthesis of cloudless areas are often considered. However, the acquisition time of different scene images varies greatly. For multiple sensors, the LST observation times are also inconsistent, and the time effect of LST affects the collaborative application of multisource LST data. The lack of temporal comparability restricts the application of polar-orbiting satellite LST data.

In recent decades, extensive efforts have focused on this field. Several approaches, such as the empirical model [25], the physical model [26], [27], [28], [29], [30], [31], [32], [33], [34], and the machine learning algorithm [24], have been developed for LST time consistency correction [1], [2]. Specifically, many of the diurnal temporal cycle (DTC) physical models have been developed [26], [27], [28], [29], [30], [31], [32], [33], which have clear physical significance and high accuracy. Liu et al. [34] applied a DTC model and Bayesian optimization algorithm for temporally normalizing LSTs to the same local solar time. They achieved orbit drift correction for long-term AVHRR data. Ma et al. [35] recently utilized the global linear model (GloLM) and a DTC model to normalize

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different LST data temporally. The results were further applied for all-weather LST estimation. The above research [34], [35], [36] has proven that these DTC models have advanced capability for LST temporal normalization. However, the DTC model can be applied only under nearly clear-sky conditions throughout the day. It is also not applicable for the conditions in which cloud contamination appears only at a part of the overpass periods.

The empirical model is therefore considered to expand the application scope. According to the statistical analysis of the linear relation between LSTs and time during 10:00–12:00 h over the morning overpass period, Duan et al. [25] established an empirical method using the relationship between the LST fluctuation slope (SLP) and the three effective parameters, including the normalized difference vegetation index (NDVI), solar zenith angle (SZA), and digital elevation model (DEM). To further consider the strong coupling effect between LSTs and other factors, Zhao et al. [24] applied a random forest (RF) model to quantify the temporal difference of the solar illumination among different local solar times and obtain the relationship between LSTs and related variables, such as solar radiation factors, vegetation, and terrain. Furthermore, they developed a dependent algorithm for temporally normalizing temporal effects to obtain a time-consistent LST product. The machine learning-based method explained the complex interaction between LSTs and related parameters. Satisfactory results were obtained when this approach was tested with MODIS data.

The DTC model exhibited limitations in the application under clear-sky conditions throughout the day. However, related studies have revealed that about 68% of the global surface is covered by clouds daily [35]. Thus, as a result, frequent cloud coverage significantly restricts the applicability of DTC models. Meanwhile, sufficient satellite measurements are required to derive the typical patterns of the diurnal LST cycle. It is difficult to derive the DTC pattern for polar-orbiting satellites for limited observations. Furthermore, when examining the LST diurnal variation curve (see Fig. 1), this was drawn from LST data of the Surface Radiation Budget Network (SURFRAD) Bondville (BND) site. It was evident that the SLP model [25] is only suitable for a narrow morning observation period (10:00–12:00 h) when LST varies linearly over time, whereas it is inadequate for extended morning periods such as 9:00–12:00 h or afternoon overpass periods like 12:00–15:00 h. Meantime, limited studies have focused on the LST variation modes specifically for these polar-orbiting satellite overpass periods when LST has nonlinear variation patterns over time. Therefore, to overcome the limitation in current research, it is crucial to develop a new general algorithm that possesses advanced capabilities to accurately characterize LST variation patterns covering broad overpass periods and various application scenarios.

Accordingly, this study introduced a hybrid and practical method integrating a new general LST variation algorithm and machine learning models. It is designed for the temporal normalization of polar-orbiting satellite-derived LST products and eliminates their temporal effects over any overpass period.

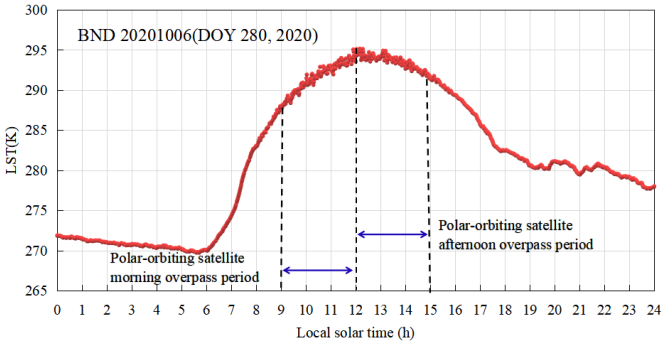


Fig. 1. Schematic for LST variation curve during clear-sky conditions only in the daytime overpass periods of polar-orbiting satellites, the SURFRAD BND site, DOY 280, 2020.

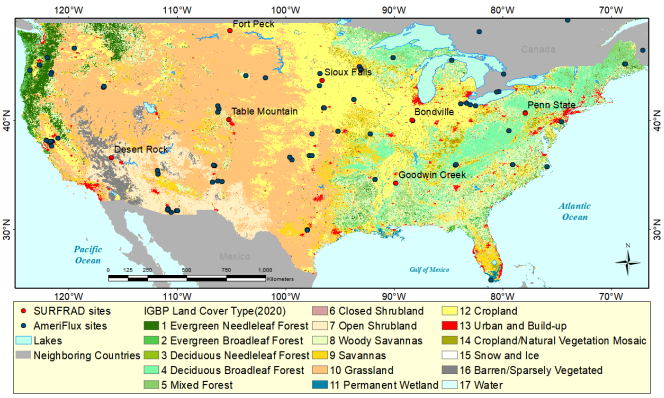


Fig. 2. Location of the study area (the contiguous United States, CONUS), the spatial distribution of used stations, and the base map showing hill-shade and IGBP LC types in 2020.

This proposed method exclusively focuses on the cloud cover condition during the observation periods, and there are no requirements accounting for weather conditions at other times, thus enhancing its applicability.

The remaining parts of this article are organized as follows. Section II introduces the study area and methods applied to acquire the data used in this research. Section III describes the methodology associated with LST variation model establishment and incorporation into the machine learning network to predict temporally normalized LSTs. Then, the results are carefully illustrated in Section IV, and the discussion in Section V helps clarify whether the proposed methodology is practical and effective. Finally, the concluding part is drawn in Section VI.

II. Study Area and Data

A. Study Area

The contiguous United States (CONUS) was selected as the study area, consisting of the 48 adjoining U.S. states and the District of Columbia on the continent of North America (see Fig. 2). The area ranges from 24°98′ N to 49°54′ N in latitude (LAT) and from −125°11′ E to −66°82′ E in longitude (see Fig. 2). Due to its large extent, this area has a wide variety of climatic zones, elevations, and land cover (LC) types. The region primarily experiences a temperate and

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TABLE I
MAJOR CHARACTERISTICS OF DATASETS USED HEREIN

Sources	Products	Annotation	Variable names	Duration	Spatial/temporal resolution
Satellite remote sensing data	MYD13A2	Normalized Difference Vegetation Index	NDVI	2003–2020	0.01°, 16-day
	MYD11A1	Land surface temperature	LST	2003–2020	0.01°, daily
	MCD12Q1	IGBP land cover types	LC	2003–2020	0.005°, yearly
	MOD15GFS	Leaf Area Index	LAI	2003–2020	0.005°, 8-day
	ASTGDEM v3	Digital Elevation Model, ASPECT, SLOPE	DEM, ASPECT, SLOPE	/	0.0003°
	GOES-R16 LST	Land surface temperature	LST	2020	0.02°, hourly
Reanalysis data	MERRA-2	Total precipitable water	TPW	2003–2020	0.5°*0.625°, hourly
		Land surface albedo	ALBEDO	2003–2020	
In-situ measurements	SURFRAD AmeriFlux	Longwave radiation, downwelling shortwave radiation	LST	2003–2020	6/30 minutes
Additional data	/	Latitude	LAT	/	0.01°
	/	Solar Zenith Angle	SZA	2003–2020	0.01°, daily

subtropical climate, characterized by moderate temperatures and ample rainfall. Such weather conditions are favorable for the flourishing of vegetation. The land use types in this area are dominated by grasslands (33.18%), croplands (17.52%), and woody savannas (12.05%).

B. Data

This study employed four types of data: 1) satellite remote sensing data; 2) reanalysis data; 3) in situ measurements; and 4) additional data. Satellite remote sensing data, reanalysis data, and other source data were utilized as the experimental datasets and validation data for temporal normalization. In situ measurements were collected to calculate the parameters of the LST temporal normalization algorithm and validation. The details of the abovementioned datasets applied in this study are presented in Table I.

1) Satellite Remote Sensing Data:

a) MODIS: The MODIS Aqua LST/Emissivity Daily L3 Global 1-km SIN Grid (MYD11A1) product from Collection 6 in 2020 was applied to test the LST temporal normalization method. The MYD11A1 LST was derived by the generalized split-window algorithm [5], [38], [39] and had been validated to have high quality and excellent accuracy.

The product includes other auxiliary information, such as the observation time, angles, emissivity, and quality control (QC). MYD13A2 is a 16-day composite grid product for the MODIS NDVI at a 1-km resolution. MCD12Q1 is a yearly MODIS International Geosphere–Biosphere Program (IGBP) LC classification scheme at a 500-m resolution. MOD15GFS is an eight-day composite dataset of MODIS combined leaf area index (LAI) and fraction of photosynthetically active radiation (FPAR) at a 1-km resolution. They were collected to obtain the NDVI, LC, and LAI data corresponding to the applied LSTs. All these datasets were stored in hierarchical data format (HDF) with sinusoidal projection. Before being applied, they were all reprojected to the World Geodetic Survey 1984 (WGS84) scheme and resampled to a spatial resolution of 0.01° by the bilinear method.

b) ASTER GDEM: Version 3 of the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) Global DEM (GDEM) was jointly released by NASA and the Ministry of Economy, Trade, and Industry (METI) of Japan. This DEM product is the most up-to-date and complete digital topographic dataset of the Earth available to the public, covering the global land surface from 83°N to 83°S LAT. It is referenced to the WGS84/EGM96 geoid. Studies on the ASTER GDEM validation had demonstrated that this product's accuracy is 20 m at 95% confidence for vertical data and 30 m at 95% confidence for horizontal data. This dataset can be downloaded from Japan Space Systems and NASA's Land Processes Distributed Active Archive Center (LP DAAC). The product was also utilized to derive the aspect and slope.

c) GOES-R16: The latest Geostationary Operational Environmental Satellite R-Series 16 (GOES-R16) has a 5-min temporal resolution and 2-km spatial granularity [40]. Based

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on its IR bands 14 (11.2 μm) and 15 (12.3 μm), a daily split-window channel emissivity proposed by the LST Algorithm Working Group at NOAA was applied to calculate the LST. The created GOES-R16 LST product has a temporal resolution of 1 h. Related validation experiments were conducted, demonstrating an accuracy below 2.5 K. This product was processed to obtain the temporally consistent LSTs at a fixed local solar time (13:30) by linear interpolation to validate the temporal normalization method. The data were reprojected to the WGS84 scheme and resampled to a spatial resolution of 0.02°.

2) Reanalysis Data: MERRA-2 is the global reanalysis developed by NASA's Global Modeling and Assimilation Office based on a common conventional and satellite observations era starting in 1980. The goals of MERRA-2 are to provide a regularly gridded, homogeneous record of the global atmosphere and to incorporate additional aspects of the climate system, including the trace gas constituents (stratospheric ozone), improved land surface representation, and cryosphere processes. This reanalysis has a horizontal grid spacing of 0.5° × 0.625° [41], [42], [43]. In this study, total precipitate water vapor (TPW) data (TQV) and land surface ALBEDO data were applied as part of the training datasets for predicting the parameters of the temporal normalization algorithm.

3) In Situ Measurements: In situ measurements were applied in this study to calculate the ground-measured LSTs at the MODIS observation times. Ground-measured longwave radiation and downwelling shortwave radiation (DWSR) are from 102 sites in the AmeriFLux and SURFRAD, covering different geographical coordinates and LC categories.

AmeriFlux was created in 1996 by many sources, including NASA, NOAA, and the U.S. Forest Service. It was a very decentralized network of PI-managed sites measuring ecosystem CO_2, water, and energy fluxes across North, Central, and South America. Its main objective is to connect field sites representing major climate and ecological biomes, including tundra, grasslands, crops, conifer, deciduous, and tropical forests [44]. The measurements are charged to the public from the following website (https://ameriflux.lbl.gov/).

The SURFRAD network was established in the mid-1990 s by the NOAA Office to support various scientific studies, such as modeling and validation studies over the United States (Augustine et al. [45]). The SURFRAD data can be freely accessed from an FTP server (ftp://aftp.cmdl.noaa.gov/data/radiation/surfrad/).

The DWSR data were utilized to filter the clear-sky measurements during overpass periods, while longwave radiation has been employed in obtaining highly accurate LST, which has found extensive applications in the evaluation of LST products and climate science research [13], [45], [46]. In this context, the ground-measured LSTs were calculated based on the Stefan–Boltzmann law

$$\mathrm{LST}\left[\frac{F^{\uparrow}-(1-\varepsilon_b)\cdot F^{\downarrow}}{\varepsilon_b\cdot\sigma}\right]^{1/4} \tag{1}$$

where $F^{\uparrow}$ and $F^{\downarrow}$are upwelling and downwelling thermal infrared (3)–50 μm) irradiance in W·m^{-2} measured simultaneously at satellite overpass periods, respectively, σ represents Stefan–Boltzmann's constant (5.670367 × 10^{-8} W·m^{-2}·K^{-4}), and ε_b refers to broadband longwave surface emissivity, assumed as a fixed value of 0.97 [47], [48].

4) Additional Data: SZA at the reference time was calculated by the solar position calculation method (http://www.srrb.noaa.gov/highlights/sunrise/solareqns.pdf) using the day of the year (DOY), the reference time, and LAT as primary input data.

III. Methodology

Fig. 3 presents the overall framework of this study. The entire LST temporal normalization process can be divided into four stages. The first stage is establishing a general LST temporal variation algorithm for any polar-orbiting satellite overpass period (see Section III-A). The second stage is obtaining the prediction models for the parameters of this general algorithm (see Section III-B). Subsequently, the third stage involves predicting the actual satellite temporal normalization parameters by utilizing the prediction models of the temporal normalization parameters and the corresponding influential variables (see Section III-C). Finally, the temporally normalized satellite-derived LSTs are calculated based on the general algorithm and the corresponding predicted actual satellite temporal normalization parameters. Also, algorithmic validation of the normalized LSTs is conducted through cross-comparisons with temporally interpolated time-consistent GOES-16 geostationary LSTs and in situ measurements (see Section III-D).

A. Establishing a General LST Temporal Variation Algorithm for Polar-Orbiting Satellite Overpass Periods

LSTs vary rapidly throughout the daytime compared with other land surface characteristic parameters, such as NDVI and LAI. In light of the diurnal variation characteristics of LSTs, the spatiotemporal patterns of LST fluctuations are significant over time (see Fig. 1). During the polar-orbiting satellite overpass periods in the morning (approximately 9:00–12:00 h) and in the afternoon (approximately 12:00–15:00 h), the clear-sky LST variation trend over time appears to have curve shapes (see Fig. 1). In addition, for a particular pixel of polar-orbiting data, the LST differences between the actual satellite observation time and the reference time (a fixed local solar time, such as 13:30 h for the afternoon overpass time) is correlated with the temporal differences between these two instantaneous times.

According to the analysis based on LSTs derived from the in situ measurements, a general algorithm was proposed for the temporal normalization of the polar-orbiting satellite LSTs. It characterizes the LST variations over the temporal differences between the actual satellite observation time and the reference time. The LST variation and temporal differences during the polar-orbiting satellite overpass periods can be expressed as follows:

$$t_\delta = t_{\mathrm{ac}} - t_{\mathrm{norm}} \tag{2}$$

$$T_\delta = T_{\mathrm{ac}} - T_{\mathrm{norm}}. \tag{3}$$

In (2), t_{ac} presents the actual specific observation time of each pixel within the MODIS data. t_{norm} indicates the reference time for normalization. t_δ is the temporal difference between

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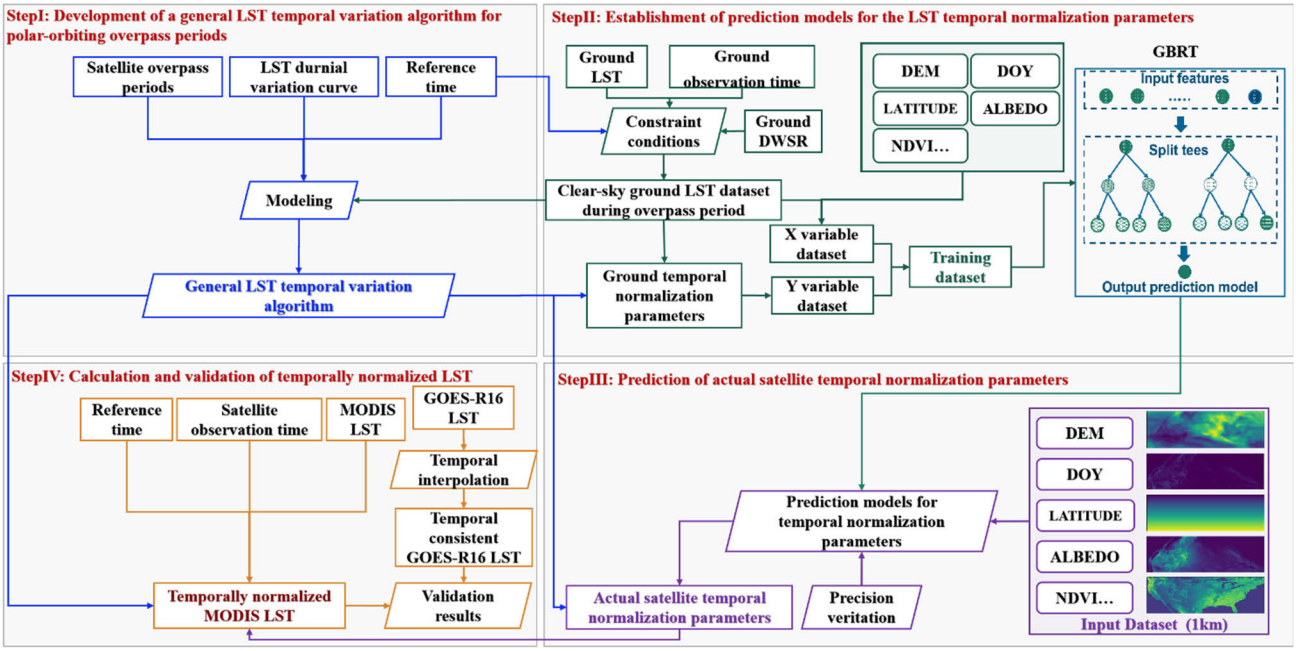


Fig. 3. Overall framework for temporal normalization of polar-orbiting satellite-derived LSTs.

the actual observation time and the reference time, which ranges with a span about from −1.5 to 1.5 h. All the time values utilized in this study are represented in local solar time. In (3), T_{ac} demonstrates the LST observed at the MODIS actual observation time t_{ac}. While T_{norm} is the predicted LST at the reference time. Then, T_δ represents the LST fluctuation between the actual observation and reference times.

As aforementioned, the polar-orbiting satellite transit periods, such as approximately 12:00–15:00 h in the afternoon and 9:00–12:00 h in the morning (see Fig. 1), the LST variation trend over time can be regarded as quadratic curves according to the LST diurnal change mode. Therefore, a general algorithm for characterizing LST temporal variation during these polar-orbiting satellite overpass periods can be expressed as follows:

$$T_\delta = at_\delta^2 + bt_\delta. \tag{4}$$

In (4), a and b are the parameters for calculating the LST variation value (T_δ) between observation and reference time. According to the analysis, when the temporal difference (t_δ) is zero, T_δ also equals zero. Thus, there is no intercept in this equation. Accordingly, the temporally normalized LSTs (T_{norm}) can be further derived as follows:

$$T_{norm} = T_{ac} - (at_\delta^2 + bt_\delta) \tag{5}$$

$$T_{norm} = T_{ac} - [a(t_{ac} - t_{norm})^2 + b(t_{ac} - t_{norm})]. \tag{6}$$

From (5), we can know that once the parameters a and b are obtained, through combining the observed LSTs (T_{ac}) and actual observation local solar time (t_{ac}) of the polar-orbiting satellite, the temporally normalized LSTs (T_{norm}) can be eventually calculated.

B. Establishing the Prediction Algorithms of LST Temporal Normalization Parameters

1) GBRT Algorithm: The gradient boosting regression tree (GBRT) [49] is an ensemble learning algorithm based on the boosting framework. The traditional regression tree models take some characteristics of the training samples as the universal properties of all samples, leading to the overfitting of the tree model. Moreover, the final output result is prone to reflecting the partial optimal solution because of using a single regression tree [50]. GBRT overcomes these shortcomings by obtaining the predicted value and residual after the initial value training and trains a series of new decision trees as models by continuously fitting the residual of the minimization loss function to the gradient-decreasing direction. Overfitting can be effectively avoided by controlling the number of iterations of the regression number, the learning rate, and the number of basic trees. GBRT's main idea is to use gradient enhancement to train each decision tree in turn in the gradient-lifting framework to achieve algorithm improvement. New regression trees are continuously constructed to better estimate the target by considering the negative gradient to minimize the loss function. In recent years, GBRT has become a versatile tool, offering high potential for applications in many fields, such as geological hazard monitoring [50], medical treatment [51], and traffic flow or speed prediction [52], [53], and geographical parameter inversions, such as latent heat flux [54] and soil moisture [55]. In this study, GBRT was utilized to obtain the two parameters of the general algorithm by decoupling atmosphere, solar radiation, and land surface features from the temporal normalization parameters.

2) Model Training:

a) Dependent variable: To obtain the dependent variables in model training, the temporal normalization parameters were calculated by the newly developed general algorithm

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TABLE II
CONSTRAINTS FOR FILTERING CLEAR-SKY MEASUREMENTS DURING POLAR-ORBITING AFTERNOON OVERPASS PERIODS

Parameters	Constraint conditions
Reference time	13:30 h
Temporal Range	12:00 h – 15:00 h
Simulation type	Quadratic polynomial
The second-order coefficient of fitted quadratic curves	<0
R	≥ 0.90
The minimum DWSR	≥ 150 W/m^2
RMSE	≤ 35 W/m^2

demonstrated in (2)–(6) and in situ measurements from the SURFARD and AmeriFlux networks. Since temporal normalization is suitable only for clear-sky conditions at the overpass periods, the in situ measurements must be selected to ensure that the data used in the study correspond to cloudless conditions during the polar-orbiting observation periods.

In this study, we took the afternoon polar-orbiting satellite data, specifically the MYD11A1 LST product from Aqua-MODIS, to test the effects of the proposed algorithm. These data's overpass period is approximately 12:00–15:00 h. Also, 13:30 h (local solar time) was set as the reference time for this polar-orbiting afternoon overpass period. After a series of tests, seven specific conditions were determined, as shown in Table II, for filtering the clear-sky in situ measurements. Analysis of the diurnal LST curve during the afternoon overpass period revealed a quadratic trend in the variation of DWSR over time, assuming cloud-free conditions. Therefore, the correlation coefficient (R) and root-mean-square error (RMSE) between the DWSR and time during this overpass period were further calculated. Subsequently, R values above 0.90 [32], [56] and RMSE values below 35 W/m^2 were used to filter the clear-sky data during the afternoon overpass periods. In addition, for clear-sky conditions, the DWSR should not be very low. According to the statistical results, DWSR values above 150 W/m^2 were subsequently utilized to filter the clear-sky data in those preliminary filtered results. Moreover, it is worth noting that among the filtered results, some simulated DWSR quadratic curves have an upward opening, which should be a downward curve, reflecting the solar radiation diurnal variations over time in the daytime. In this case, the second-order coefficient of the curve is set to be less than zero. These constraints can help filter out the clear-sky data during the polar-orbiting afternoon transit time to the greatest extent.

Based on the filtered clear-sky in situ LST data, the proposed general algorithm for temporal normalization was applied to calculate the temporal normalization parameters. Those data were viewed as dependent variables of training datasets for establishing the parameter prediction models. In the process, the LSTs at the reference time (13:30 h in local solar time) were interpolated by the linear method using the two or four measured LSTs in 1 h. Then, the simulation results with R values above 0.95 are selected. Fig. 4 shows the histograms of the temporal normalization parameters a and b extracted from the filtered clear-sky in situ LST data. The span

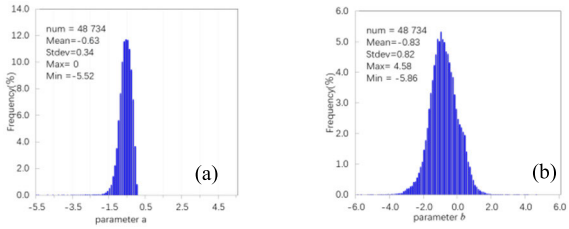


Fig. 4. Histograms of temporal normalization parameters from filtered clear-sky in situ measurements, (a) and (b) are for parameters a and b, respectively.

ranges are -5.52 to 0 and -5.86 to 4.58 for parameters a and b, respectively.

b) Independent variables: Selecting the proper independent variable combination is crucial for LST temporal normalization. LST spatiotemporal pattern is closely related to land surface characteristics, atmosphere, and solar radiation. Considering the parameter variation over different conditions, we initially selected the following variables: DOY, TPW, LAI, NDVI, SZA, LAT, LC, DEM, SLOPE, ASPECT, and ALBEDO. First, LST exhibited different periodic changes for complicated influences of multiple factors, which can be reflected by the DOY variable. In addition, LST generally decreases with elevation, and aspect and slope also affect thermal radiation distribution. Furthermore, different LC types can absorb or reflect various solar radiation, causing changes in LST. NDVI and LAI can represent vegetation coverage. Meantime, a strong correlation and intricate interplay exist between LST and TPW, involving cloud formation and other influences. While ALBEDO controls the proportion of solar radiation reflected into the atmosphere from the surface, it would indirectly affect LST changes. In addition, SZA and LAT are also close to solar radiation.

To mitigate overfitting, we implemented a feature selection process in our GBRT models for predicting temporal normalization parameters. We carefully chose the most crucial variables and adjusted parameters accordingly. The mean decrease in impurity (MDI) index, a widely used method in tree models, was employed to measure variable importance due to its efficiency and stability. This index quantifies the reduction in node impurities achieved by splitting on a specific variable across all trees, thereby indicating its contribution to the model. After conducting the feature selection, we identified the variable LC as the least significant and subsequently removed it from consideration. During our analysis, we noticed that both LAI and NDVI exhibited similar land surface vegetation cover conditions. Consequently, removing LAI did not have a substantial impact on the results. Finally, the prediction models f_a and f_b for LST temporal normalization parameters a and b are expressed as follows, respectively:

$$\text{Coeff}_a = f_a(\text{DEM, SLOPE, ASPECT, NDVI, LAT, DOY, TPW, ALBEDO, SZA}) \quad (7)$$

$$\text{Coeff}_b = f_b(\text{DEM, SLOPE, ASPECT, NDVI, LAT, DOY, TPW, ALBEDO, SZA}). \quad (8)$$

After filtering the invalid data about these nine features, the training datasets for parameters a and b eventually comprise

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48 735 clear-sky samples. These data covered 95 sites, and the temporal range spans from 2003 to 2020, utilizing DOY throughout the year. Most conditions over CONUS regarding geography diversity, underlying surfaces, latitude, elevation, and climatic conditions are covered by these datasets, as shown in Fig. 2. These stations are capable of covering all conditions across CONUS.

In the training and testing process, the samples were randomly split into a training set and a test set at a ratio of 8:2. Several important GBRT parameters were adjusted as follows: n_estimators = 300, max_depth = 4, min_samples_split = 5, learning_rate = 0.1, and loss = squared_error.

C. Prediction of the Temporally Normalized LSTs

After the first two stages, the general algorithm in (2)–(6) and the prediction models of the temporal normalization parameters in (7) and (8) were obtained. Subsequently, it was to predict the actual temporal normalization parameters of MODIS LST data in Stage III (see Fig. 3).

This required preparing the influential dependent variables for the prediction datasets, similar to the independent variables in the training datasets in Section III-B2. The variables in the training datasets were point-scale data extracted from the prepared dataset in Table I, corresponding to the filtered clear-sky in situ LST data; the utilized variables in the prediction dataset were image data corresponding to the satellite LST product. The data source details can be accessed in Table I. All the data were reprojected to WGS84 and resampled to a spatial resolution of 0.01°.

Subsequently, after inputting the prediction variable dataset into the parameter prediction models from Stage II, the actual temporal normalization parameters for MODIS data during the overpass periods can be obtained. Finally, as shown in Stage IV (see Fig. 3), the temporally normalized LSTs can be calculated using the empirical algorithm, the actual observed LSTs, and the predicted actual satellite parameters.

D. Evaluation Strategies

Comprehensive validation of the temporally normalized MODIS LST data was required before subsequent applications. An independent validation was conducted by cross-comparisons between the temporally normalized MODIS LST and the stationary satellite GOES-R16 LST data over the CONUS. Before comparisons, it needs to calculate the time-consistent GOES-R16 LSTs at the reference time with a linear interpolation method. First, we chose the GOES-R16 LSTs during the afternoon overpass period (12:00–15:00 h) and converted the original UTC to local solar time. Then, we set the reference time as 13:30 h (local solar time) and utilized a linear interpolation method to interpolate the temporally adjacent 2–4 observations around the reference time, resulting in time-consistent GOES-R16 LSTs. Finally, we compared these data with our results to confirm effective validation.

In addition, it is well known that the time and angle effects were coupled in the LST products. Thus, it was optimal for reducing the influence of the angle effect in the comparison process. To minimize the angular discrepancies between MODIS and GOES-R16 LST data, pixel pairs with VZA angle differences less than 15° and VZA angles less than 50° were used. In addition, the temporally normalized MODIS LST data were aggregated from 0.01° to 0.02° to be compatible with the GOES-R16 LST data by the bilinear method. The indicators, including correlation coefficient (R), RMSE, mean absolute error (MAE), standard deviation (STD), and bias, were applied to quantify the temporal normalization results. The validation process was divided into two temporal differences (0–1 h and >1 h) due to constraints imposed by the VZA limitations, as the majority of temporal differences between observation and reference time fell within the 1-h range. Different analytical strategies were employed for each range to accommodate the availability of time series data. Within the 1-h range, comprehensive validation was performed, considering different temporal differences (TDs) including 0–1.0, 0–0.5, and 0.5–1 h, through cross-comparisons across individual days, time series, seasons, and LC types. Moreover, to improve the accuracy and reliability of the proposed method, seven SURFRAD sites were chosen as validation site data due to their exceptional observation precision and also because they are dispersed among different underlying surfaces, climatic conditions, geographical diversity, and elevations throughout the study region, while, for TDs exceeding 1 h, results on individual days were mainly focused, with the threshold of VZA differences increased to 20° for filtering more pixel pairs.

IV. Results

A. Validation Results for Various Temporal Differences Within 1 h

1) Model Performance on Individual Days: Fig. 5 shows the density scatter plots of the original and the normalized LSTs versus the interpolated time-consistent GOES-16 LSTs at 13:30 h. Overall, the results indicate that the accuracy of the temporally normalized LSTs is higher than that of the original LSTs. In detail, for TDs of 0–1.0 h, R on 2020 114 and 2020 297 was improved by approximately 0.2 (from 0.85 to 0.87) and 0.1 (from 0.93 to 0.94), respectively. Furthermore, the proposed algorithm also improved the RMSEs from 1.86 to 1.67 K (DOY 009), 3.00 to 2.85 K (DOY 114), 1.67 to 1.57 K (DOY 279), and 2.41 to 2.27 K (DOY 297); correspondingly, the RMSEs decreased by approximately 0.10, 0.15, 0.09, and 0.15 K, respectively (see Fig. 5). At the same time, the normalized LSTs also had lower bias values than the original LSTs. The bias values of the normalized LSTs were only −0.14, −0.19, and −0.03 K at DOY 2020009, 2020274, and 2020297, respectively. A similar trend could also be found from the validation results with TDs of 0–0.5 h. The validation results suggest that the accuracy showed a slight improvement when the TDs were between 0 and 1 h, as well as between 0 and 0.5 h, while, for the validation results with TDs of 0.5–1.0 h in Fig. 5, the RMSEs after normalization were improved from 2.33 (before normalization) to 1.39 K at DOY 009, 3.58 (before normalization) to 2.72 K at DOY 114, 2.31 (before normalization) to 1.55 K at DOY 274, and 2.69 (before

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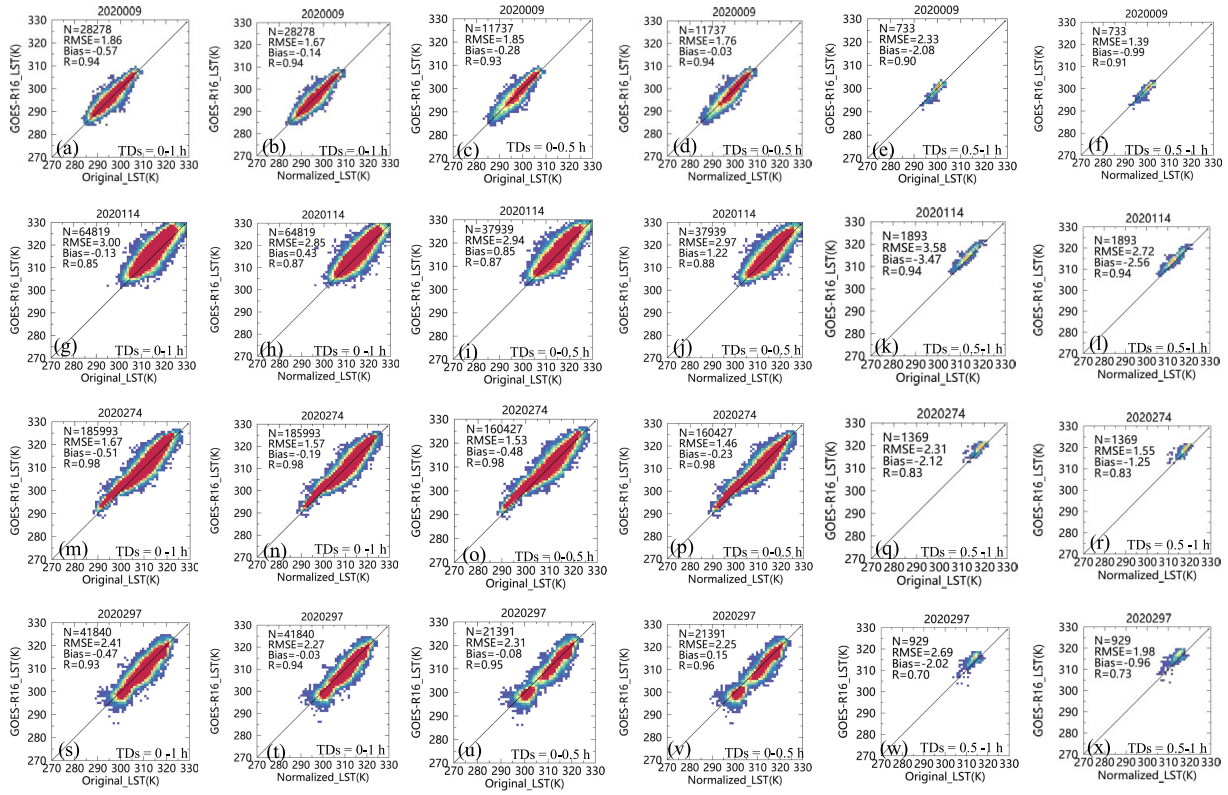


Fig. 5. Density scatter plots between the temporally normalized LSTs versus GOES-R16 LSTs and original LSTs versus GOES-R16 LSTs over the CONUS for different TDs. (a)–(f) January 9, 2020 (DOY 009). (g)–(l) April 23, 2020 (DOY 114). (m)–(r) September 30, 2020 (DOY 0274). (s)–(x) October 23, 2020 (DOY 297). TDs indicate the temporal differences, which are measured in hour.

normalization) to 1.98 K at DOY 297. The RMSEs improved are about 0.7–1 K.

In addition, Fig. 6 gives the histograms of the LST differences between the MODIS LSTs and GOES-R16 LSTs with TDs of 0.5–1.0 h. In the histograms of the normalized LSTs, all the columns shifted closer to the value of 0 compared to the histograms before normalization. In detail, for the mean error (mean) (after normalization), the values (before and after normalization, respectively) were −2.08 and −0.99 K at DOY 009, −3.47 and −2.56 K at DOY 114, −2.12 and −1.25 K at DOY 274, and −2.02 and −0.96 K at DOY 297. From the statistics with TDs of 0.5–1.0 h, one can easily conclude that the validation results are substantially improved.

2) Model Performance for Annual Time Series: Fig. 7 shows the RMSE differences (RMSEs before normalization minus those after normalization) for the annual time series 2020 with various TDs. For the results with TDs of 0–1.0 h in Fig. 7(a) and TDs of 0–0.5 h in Fig. 7(b), the differences between the RMSEs of the original LSTs versus GOES-LSTs and RMSEs of normalized LSTs versus GOES-LSTs in 2020 were −0.15 to 0.30 K and −0.24 to 0.49 K, respectively. This indicates that normalization results were slightly superior to the original ones.

The results in Fig. 7(c) for 0.5–1.0 h had RMSE differences ranging from −0.06 to 1.03 K. RMSE differences in the whole year above 0.5, 0.7, and 0.9 K are approximately 50%, 71.57%, and 89.22%, respectively. Some negative RMSE difference values could be found in the histograms, but the percentages are few. They are about 2.96%, 14.92%, and 1%, accounting for all 338, 295, and 101 tests for TDs of 0–1.0 h, 0–1.5 h, and 0.5–1.0 h, respectively. Fig. 8 shows the annual series for the RMSE and bias differences (indicators of original MODIS LSTs minus those of the normalized LSTs) with temporal differences of 0.5–1 h. The results in Fig. 8 indicate that accuracy after normalization was substantially improved.

Table III lists the statistical results of original and normalized LSTs and their differences for the time series in 2020 with TDs of 0.5–1.0 h. The RMSE and bias differences span from 0.320 to 1.034 K and from 0.303 to 1.267 K, respectively. From this multiple statistical analysis during various TDs, neither the RMSEs nor the bias values varied significantly with the portion of the relatively considerable temporal differences. In general, the accuracy of the results after normalization with TDs = 0.5–1.0 h exhibited more substantial improvements than the other TDs.

3) Model Performance in Each Season: Fig. 9 gives the validation results of the proposed algorithm in each season. For the total statistical values, the results in spring and summer, both after normalization and before normalization, had lower accuracies than those in other seasons at all TDs (0–1.0 h, 0–0.5 h, and 0.5–1.0 h). For example, the RMSEs of results with TDs of 0.5–1.0 h in spring (the RMSEs of the results before and after normalization were approximately equal to 3.9 and 3.3 K, respectively) and in summer (the RMSEs of the results before and after normalization were approximately equal to 4.1 and 3.7 K, respectively) were higher than those in autumn (the RMSEs of the results before and after normalization were approximately equal to 3.0 and 2.5 K,

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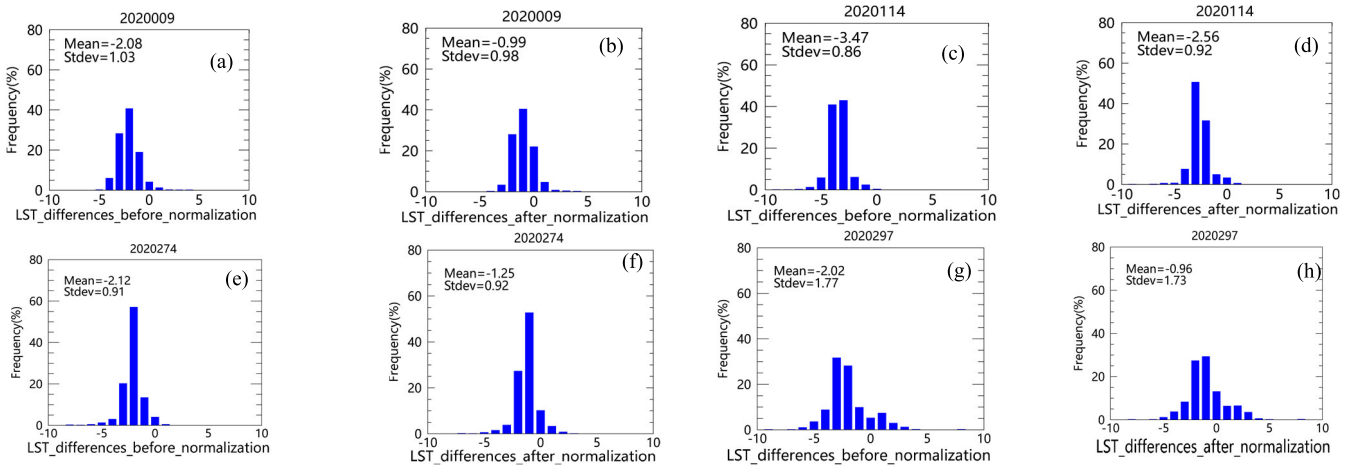


Fig. 6. Histograms of LST differences (the original and the normalized LSTs minus the corresponding time-consistent GOES LSTs) for individual days with temporal differences between 0.5 and 1.0 h, (a)–(b) January 9, 2020 (DOY 009), (c)–(d) April 23, 2020 (DOY 114), (e)–(f) September 30, 2020 (DOY 0274), (g)–(h) October 23, 2020 (DOY 297). LST differences were measured in kelvin (K).

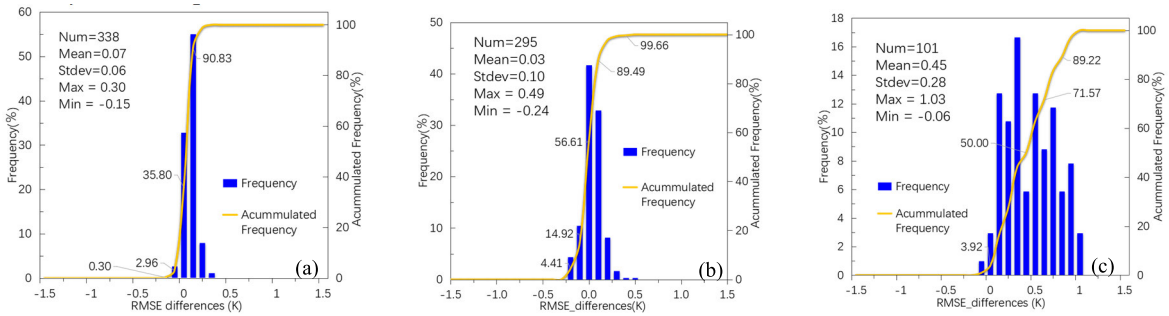


Fig. 7. Histograms of RMSE differences (RMSEs before normalization minus RMSEs after normalization) with (a) TDs = 0–1.0 h, (b) TDs = 0–0.5 h, and (c) TDs = 0.5–1.0 h; the unit of LST is kelvin (K). TDs indicate the temporal differences, which are measured in hour.

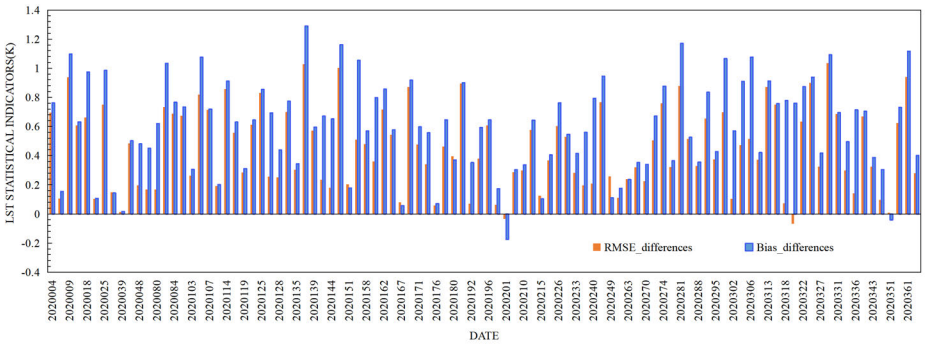


Fig. 8. Annual series for the RMSE and bias differences (indicators of original LSTs minus those of the normalized LSTs) with temporal differences of 0.5–1 h; RMSE and bias values are measured in kelvin (K).

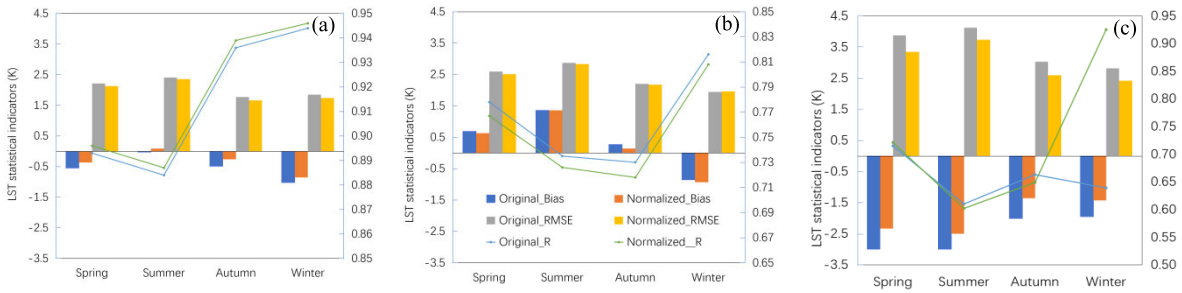


Fig. 9. Statistical indicators (RMSE, bias, and R^2 values) of validation results in each season with different TDs. (a) 0–1 h. (b) 0–0.5 h. (c) 0.5–1.0 h. RMSE and bias values were measured in kelvin (K).

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TABLE III
PART OF THE VALIDATION RESULTS IN TIME SERIES WITH $T_{DS} = 0.5$–1.0 h

Date	Samples	RMSE (K)			Bias (K)		
		Original	Normalized	Differences	Original	Normalized	Differences
2020004	70	2.215	1.519	0.696	-2.133	-1.355	0.778
2020091	56	9.274	8.757	0.517	-8.544	-8.155	0.389
2020098	281	3.905	3.231	0.674	-3.622	-2.905	0.717
2020125	28	4.846	4.016	0.830	-4.703	-3.833	0.870
2020137	1028	3.495	2.467	1.028	-2.811	-1.544	1.267
2020146	141	3.669	2.667	1.002	-3.425	-2.311	1.114
2020169	191	5.110	4.239	0.871	-4.087	-3.225	0.862
2020187	32	6.641	5.745	0.896	-5.902	-5.046	0.856
2020267	99	2.723	2.403	0.320	-2.652	-2.349	0.303
2020281	917	3.191	2.312	0.879	-2.503	-1.370	1.133
2020304	467	4.616	4.144	0.472	-2.346	-1.530	0.816
2020324	73	2.435	1.536	0.899	-2.374	-1.474	0.900
2020329	229	3.152	2.118	1.034	-2.906	-1.860	1.046
2020361	653	2.388	1.448	0.940	-2.127	-0.993	1.134

respectively) and winter (the RMSEs of the results before and after normalization were equal to 2.8 and 2.4 K, respectively).

Despite this tendency in the total values of indicators, no prominent seasonal characteristics could be found regarding the indicator variations between the normalized and original LSTs. The accuracies of all LSTs after normalization were improved over the original LSTs to a certain extent without evident differences. In addition, the indicators (RMSE and bias) after normalization with TDs of 0.5–1.0 h invariably presented more substantial improvements than those with TDs of 0–0.5 and 0–1.0 h. The R values obtained at TDs of 0–1.0 h in each season were also improved. In particular, the R values after normalization with TDs of 0.5–1.0 h in winter were approximately improved to 0.925 from 0.639 (before normalization).

4) Model Performance in Various Land Cover Types: Fig. 10 shows the statistical validation results over various LC types for TDs of 0–1.0 h for all LC types. Regarding the correlation R results, there were no substantial changes in the coefficient values for most LC types. For the bias values, barren or sparsely vegetated and open shrublands showed the most remarkable differences between the normalized and original LSTs, reaching approximately 0.4 K, while the bias variation after the normalization of the remaining LC types fluctuated between 0.1 and 0.32 K. The RMSE values consistently improved after performing the normalization process, resulting in higher accuracy. Specifically, for the deciduous needle leaf forest, the RMSE was reduced from 2.36 (before normalization) to 2.13 K (after normalization), indicating a notable enhancement in accuracy. Table IV shows the validation results against in situ measurements, which revealed that inherent systematic errors exist between the raw and station data. These errors span from approximately −6.54 to 5.18 K, while compared to these values, the discrepancies between LST differences from MODIS and in situ datasets were minor, with some of them being almost identical.

The proposed algorithm effectively improved the accuracy of the results compared to the prenormalization stage. Moreover, the study found consistent and reliable outcomes across various LC types, emphasizing the practicality and viability of the proposed approach for LST normalization.

B. Validation Results for Temporal Differences Above 1 h

The validation results with temporal differences above 1.0 h at DOY 077 are shown in Fig. 11. The TD span was −1.7 to 1.0 h. The normalized RMSE decreases from 4.17 (before normalization) to 3.89 K. The mean LST difference (bias) changed from −3.62 (before normalization) to −1.92 K (after normalization). According to the plots at DOY 077, the centerline (red dashed line) of the scatter for the normalized LSTs in Fig. 11(b) moved obviously toward the 1:1 line (solid black line) in contrast with the centerline (red dashed line) of the original LSTs in Fig. 11(a). This phenomenon corresponds to the bar plot distribution. Similarly, from the analysis of Fig. 11(d) and (e), the bars of the normalized LSTs also evidently moved closer to 0. Correspondingly, the LST differences after normalization have lower mean values than those before normalization.

V. DISCUSSION

A. Strengths

1) Analysis of Simulating LST Temporal Variations: As the modeling process in Section III-A, (6) was finally utilized for LST normalization. It was derived from (4), which captures LST temporal variation characteristics by simulating relationships between the LST differences and temporal differences specifically under clear-sky conditions during the overpass period. As such, its accuracy is largely determined by the simulation accuracy. Thus, we have provided scatter plots that show the simulation results based on station-derived LST data with a high temporal resolution, in addition to histograms of

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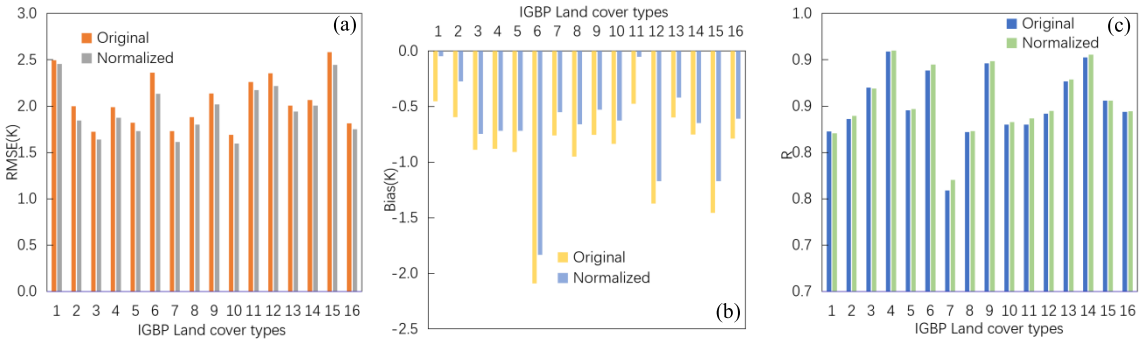


Fig. 10. Comparisons of statistical indicators: (a) RMSE (K), (b) bias (K), and (c) R in various IGBP LC types (1)—evergreen needle leaf forest; 2—evergreen broadleaf forest; 3—deciduous needle leaf forest; 4—deciduous broadleaf forest; 5—mixed forests; 6—closed shrublands; 7—open shrublands; 8—woody savannas; 9—savannas; 10—grasslands; 11—permanent wetlands; 12—croplands; 13—urban areas; 14—cropland natural vegetation mosaic; 15—snow and ice; and 16—barren or sparsely vegetated).

TABLE IV
COMPARISON RESULTS WITH IN SITU LSTS

Site names	Date	MODIS OLST(K)	MODIS NLST(K)	In-situ OLST(K)	In-situ NLST(K)	MODIS LST differences(K)	In-situ LST differences(K)	Temporal differences(h)
BND	2020150	306.14	306.07	301.43	299.49	0.07	1.94	-0.30
	2020199	302.84	302.40	307.11	303.37	0.44	3.75	0.40
DRA	2020026	292.62	293.16	295.41	295.39	-0.54	0.02	-0.40
	2020079	291.34	290.20	296.52	291.77	1.14	4.75	-0.60
FPK	2020065	282.96	282.93	283.38	282.64	0.03	0.74	-0.10
	2020208	310.82	311.01	307.74	307.90	-0.19	-0.17	-0.80
PSU	2020146	300.02	301.52	302.05	303.20	-1.50	-1.15	-0.90
	2020174	302.54	302.74	302.83	305.77	-0.20	-2.94	-0.50
SXF	2020112	305.74	304.71	301.12	300.29	1.03	0.83	-0.20
	2020281	300.30	300.04	299.42	298.55	-0.26	-0.87	-0.90
TBL	2020167	317.16	316.82	321.56	321.07	0.34	0.49	-0.70
	2020169	317.66	317.34	322.49	322.05	0.32	0.44	-0.90
GWN	2020131	297.54	297.23	304.08	304.03	0.31	0.04	-1.00
	2020202	310.24	310.07	309.78	309.90	0.17	-0.13	-0.90

Notion: In-situ OLST and NLST indicate measured LST at the original observation time and reference time, respectively; the LST differences demonstrate LST at MODIS actual observation time minus LST at the reference time. Bondville (BND), Goodwin Creek (GWM), Penn State (PSU), Sioux Fall (SFX), Table Mountain (TBL), Desert Rock (DRA), Fort Peck (FPK).

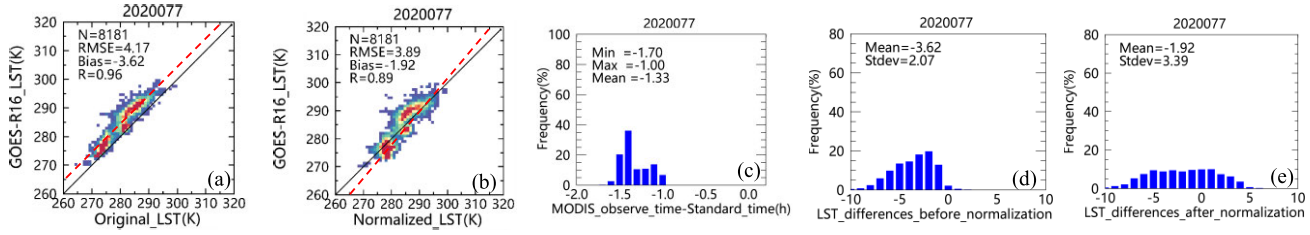


Fig. 11. Scatter plots of the time-consistent GOES-R16 LSTs versus (a) original LSTs and (b) normalized LSTs, (c) histograms of temporal differences (MODIS observation time minus the reference time), and histograms of LST differences (d) before and (e) after normalization, with original LSTs and normalized LSTs minus GOES-R16 LSTs, respectively.

RMSE and MAE values for all the samples utilized in the training datasets (see Fig. 12). The excellent simulation performances fully demonstrated that (6) can accurately capture the LST variation patterns at any overpass period under clear-sky conditions. Thus, the proposed algorithm has the advantage of high fitting accuracy.

2) Analysis of Overall Statistical Features: According to the quantitative evaluation results in Section IV, the proposed LST temporal normalization algorithm provided reliable results for all seasons and various LC types at large temporal and spatial scales. The analysis of the total statistical results in each season shows that the spring and summer had lower accuracies between the MODIS LSTs and GOES-R16 LSTs than the other two seasons. The normalized LSTs presented a similar seasonal trend to the original LSTs regarding the total values. This seasonal phenomenon was also found in the research from [57]. The seasonal characteristics were attributed to the fact that vegetation's growing

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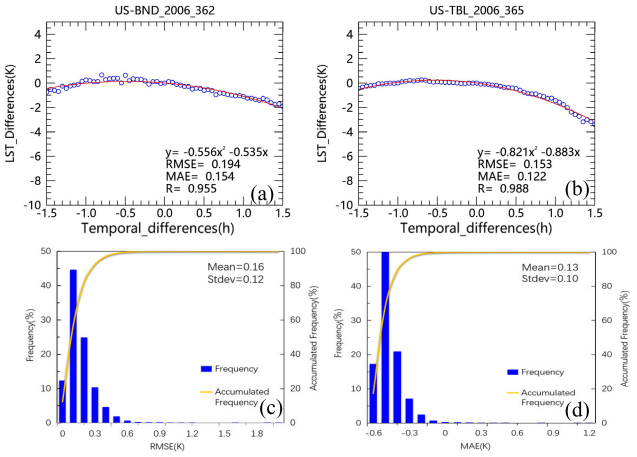


Fig. 12. Examples of scatter plots between the LST differences (the original LST minus the LST at reference time, unit: K) and temporal differences (observation time minus the reference time, unit: h) under clear-sky condition during the afternoon overpass period (12:00 h–15:00 h). (a) SURFRAD BND site, DOY 362, 2020. (b) SURFRAD TBL site, DOY 365, 2020. Histograms of (c) simulation RMSE and (d) MAE values for samples utilized in training datasets.

and peak seasons could significantly impact the LST values. Although seasonal tendencies exist in the total statistical value results, there is no significant difference in normalization performance between different seasons within the same TD range. This demonstrates that the proposed algorithm has advanced applicability and stability for all application scenarios. Regarding the station validation results covering various LC types, significant systematic errors between raw and station data were exhibited. The uncertainty could arise from inaccurate cloud detection, in situ site heterogeneity, terrain effect, or algorithmic uncertainty [58]. However, the LST difference values displayed minor discrepancies, which reduced uncertainty during comparison, while, for results compared with GOES-R16 LST, the normalized LSTs showed a comparable level of enhanced accuracy across different LC types than before normalization. The key factors influencing the performance at different LC types lie in that the nine selected features can adequately capture parameter variation patterns for normalization, making LC types less crucial in parameter prediction. Therefore, the training datasets can cover all the LC conditions, which ensures prediction accuracy over various LC types.

A phenomenon worth noting was observed in Fig. 5 of 2020114, where bias decreased for temporal differences less than 0.5 h. Compared to the larger temporal differences above 0.5 h, these results within 0.5 h had a more considerable impact on the bias changes. After analyzing the data, we found that this phenomenon can be attributed to two factors. First, LST variation was typically minimal during these short temporal differences within 0–0.5 h. Second, there were data errors present in both MODIS and GOES LST datasets. Therefore, data within 0.5 h are likely to have more influence on the decrease or increase of bias values between prior and after normalization. In addition, very few negative RMSE difference values were found in Fig. 7. This is also mainly caused by the inversion errors from the original GOES-R16 and MODIS LST products. Some of the results shown in Fig. 8(b) and (c) are also notable. The R results in some seasons with TDs of 0.1–0.5 h and 0.5–1.0 h did not have higher accuracies than the original ones, as expected. After analysis, this phenomenon is because the samples in these cases are very few, and then, the potential errors from LST products may substantially influence the final seasonal R coefficient. Moreover, the validation results showed an evident tendency that the accuracies with larger TDs were improved more substantially than those with small ones. For TDs of 0–0.5 h, there were slight differences between the validation results of the normalized and original LSTs. Regarding the results with TDs of 0.5–1.0 h, the RMSEs and mean LST differences between the original and normalized results showed a relatively substantial decrease on most individual days. The RMSE and bias improvements reached 1 K among the validation results.

3) Comparison With the Current Research: The study proposed an improved comprehensive approach for temporal normalization of polar-orbiting LSTs, which combines a newly developed general algorithm and ensemble regression models. When tested with the MODIS MYD11A1 LST data, the normalized LSTs showed higher accuracies than the original LSTs. Compared to the current research, the proposed method possesses certain advantages.

The proposed algorithm is more broadly applicable than traditional physical models. In particular, the DTC physical model is suitable only for clear-sky conditions throughout the day and usually requires at least four observed LST values. Thus, the DTC model alone is inapplicable for temporal normalization of polar-orbiting satellite-derived LST. The algorithm proposed in this study only needs to fulfill the condition of being clear-sky during the satellite transit period. Moreover, there are only two parameters in the model. In addition, in comparison with the SLP empirical method developed by Duan et al. [25], the proposed method is improved in two ways. On the one hand, the general algorithm was improved from an SLP linear function to a nonlinear function suitable for any morning and afternoon overpass periods. It is worth noting that when applying the proposed temporal normalization general algorithm during the morning period of 10:00 h–12:00 h, because of the linear trend of LST variation over time, the quadratic parameter a would be equal to 0 for this period. On the other hand, we upgraded the temporal normalization parameter prediction method from three simple variable linear functions to nonlinear relationships between nine effective variables, involving land statistics, atmospheric, and solar radiation. The method developed herein integrates the advantages of the general empirical algorithm and machine learning models to reduce its sensitivity to data errors and improve its accuracy and robustness.

B. Limitations

Although the proposed algorithm shows high accuracy and efficiency, some limitations can still affect the temporal normalization performance. To establish the parameter prediction models, LSTs derived from measurements of 95 sites were utilized to extract the temporal normalization parameters as

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independent variables in the training datasets. In the process, the in situ LST representativeness was not considered. As described in Section III-B2, the LST data in the training datasets were derived from in situ measurements at a point scale, whereas the other variables in the training and prediction datasets were obtained from satellite images at a 0.01° resolution or reanalysis data at a coarser resolution. Generally speaking, measurements from sites with homogeneous characteristics have good representativeness for the entire pixels. However, to ensure that sufficient sites distribute all across the study area, the sites in the study were not filtered by initial selection according to the heterogeneity. Some heterogeneous sites many existed in the utilized sites. Therefore, the point-scale LST data and corresponding extracted normalization parameters from these heterogeneous sites may not be able to represent the entire pixels. Thus, these scale effects among various datasets and spatial heterogeneity likely led to some uncertainties in the normalization results.

In addition, the time effect is coupled with the angle effect in LST products, and the accuracy of the temporally normalized LSTs is closely related to the change in the VZAs. After normalization, the overall results with temporal differences above 1 h show an improved accuracy. It is undeniable that there are still some VZA effects that may affect the validation results because of the relatively broader VZA restraint conditions.

C. Prospects

The proposed method herein has a high temporal normalization efficiency. Once trained, it can efficiently generate long-time-series temporally normalized LST with time consistency. These generated products can be applied in agricultural disaster monitoring, climate change analyses, heat island effect monitoring, and evapotranspiration research.

As mentioned, this proposed approach is suitable for any polar-orbiting satellite overpass period. Thus, it can also be an alternative method for the temporal normalization of MODIS LST products during morning overpass times. The application to LSTs during morning overpass periods will be performed in the near future. In addition, we are also striving to improve the temporal normalization performance by achieving scale consistency among the training and prediction datasets. Instead of relying on point-scale in situ measurements, one potential alternative to investigate novel solutions for temporal normalization of polar-orbiting satellite LST products is utilizing geostationary satellite LST products, which offer high temporal resolutions, as well as similar spatial resolutions with MODIS LST products.

VI. Conclusion

Due to the observed characteristics of polar-orbiting satellite sensors, the pixels along the scan lines are observed at different solar local times. The lack of practical temporal normalization methods suitable for LSTs observed at any polar-orbiting overpass period limits the usage of data collected under the conditions of large areas and multiple satellites. A new general LST temporal normalization algorithm is highly required to conduct temporal correction for LSTs at any satellite transit period to obtain LST data at a consistent local solar time.

This study aims to investigate the potential use of a comprehensive temporal normalization approach that integrates a novel general LST variation algorithm and ensemble regression models. The method was designed to be applicable for polar-orbiting satellite-derived LSTs at any overpass period. As long as there are clear-sky conditions during the polar-orbiting overpass periods, irrespective of the weather conditions at other times of the day. The LST temporal variation algorithm was established by characterizing the LST variations over temporal differences between the observation and the reference time, which only have two parameters and high simulation accuracy. Furthermore, prediction models for the parameters were obtained by invoking GBRT models by establishing the relationships between these parameters and nine effective factors. The training datasets were created by utilizing in situ measurements to derive temporal normalization parameters as dependent variables and collecting sufficient features acting on LST as independent variables. This approach was tested with the instantaneous Aqua-MODIS LST product (MYD11A1) in 2020 over the CONUS. Moreover, the results were validated with temporally interpolated time-consistent GOES-R16 LST and SURFRAD site data, covering various time ranges across each season and different LC types. It was found that temporally normalized LSTs have higher accuracy than the original MODIS LSTs, indicating that the proposed method possesses a good potential for effectively mitigating the temporal effect of wide-view polar-orbiting satellite observations. Thus, this work provides a new general solution with advanced capabilities for the temporal normalization of polar-orbiting satellite-derived LST products.

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